

PS C:\Users\aswini\OneDrive\Desktop\wordpiece_tokenizer> python wordpiece.py

```
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WORDPIECE TOKENIZER
=====
```

Training words:

```
{'cat': 3, 'cats': 2, 'dog': 1}
```

Initial splits:

```
cat -> ['c', '##a', '##t']
```

```
cats -> ['c', '##a', '##t', '##s']
```

```
dog -> ['d', '##o', '##g']
```

Token frequencies:

```
c = 5
```

```
##a = 5
```

```
##t = 5
```

```
##s = 2
```

```
d = 1
```

```
##o = 1
```

```
##g = 1
```

Pair frequencies:

```
('c', '##a') = 5
```

```
('##a', '##t') = 5
```

```
('##t', '##s') = 2
```

```
('d', '##o') = 1
```

```
('##o', '##g') = 1
```

WordPiece scores:

```
('c', '##a') = 0.2
```

```
('##a', '##t') = 0.2
```

```
('##t', '##s') = 0.2
```

```
('d', '##o') = 1.0
```

```
('##o', '##g') = 1.0
```

Best pair:

```
('d', '##o')
```

Best score: 1.0

```
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```
('d', '##o') = 1.0  
('##o', '##g') = 1.0
```

```
Best pair:  
('d', '##o')  
Best score: 1.0
```

```
New token created:  
do
```

```
Updated splits:  
cat -> ['c', '##a', '##t']  
cats -> ['c', '##a', '##t', '##s']  
dog -> ['do', '##g']
```

```
Vocabulary:  
{'c', '##t', '##s', 'do', '##o', 'd', '##a', '##g'}
```

```
Tokenization:  
dog -> ['do', '##g']
```

```
Tokens:  
['do', '##g']
```

```
Token IDs:  
[8, 7]
```

```
Unknown word test:  
xyz -> ['[UNK]']
```

```
=====
```

FINAL RESULT

```
=====
```

```
Input word : dog  
Tokens      : ['do', '##g']  
Token IDs   : [8, 7]
```

```
WordPiece process completed!
```